

OMAR KHATTAB

Frontend Developer (React / Next.js / TypeScript) | Engineering Software Developer

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[Portfolio](#) [GitHub](#) [LinkedIn](#)

SUMMARY

SUMMARY Mechanical Power Engineer turned **Engineering Software & Frontend Developer** with expertise in **React, Next.js, TypeScript, and Tailwind CSS**. Bridges the gap between complex thermodynamic/fluid mechanics equations and high-performance web applications. Specialized in building interactive engineering calculators, MEP software tools, and dynamic dashboards. Available immediately for Junior Frontend Developer and Engineering Software roles in Cairo.

TECHNICAL SKILLS

Frontend Architecture: React.js, Next.js, TypeScript, JavaScript ES6+

UI & Styling: Tailwind CSS, Material UI, Framer Motion, CSS Modules

State Management: React Hooks, Context API, Form Validation

APIs & Tools: REST APIs, Git, GitHub, Vercel, CI/CD

Engineering Software: AutoCAD, Autodesk Inventor, ANSYS,

Web Development: Responsive Design, Flexbox, CSS Grid, Semantic HTML5

FEATURED PROJECTS

HVAC Engineering Suite & Calculators [Live Demo](#) | [Code](#)

Sep 2026 – Present

- Developed a web application for MEP engineers to calculate cooling loads, supply-air CFM, and duct dimensions in real time.
- Implemented **duct-sizing algorithms** using **Equal Friction and Equal Velocity methods**.
- Added aspect-ratio constraints and circular-to-rectangular equivalent-diameter calculations.
- Built a **thermal-load module** covering envelope conduction, solar gain, occupancy heat, lighting loads, and equipment loads.
- Integrated multi-unit conversions for flow rates, pressure losses, and cooling capacity.

E-Commerce Store [Live Demo](#) | [Code](#)

Jul – Aug 2026

- Built a complete e-commerce storefront using React, React Router, and Context API.
- Integrated live product REST APIs with loading states and error handling.
- Implemented cart persistence using localStorage.
- Developed a mobile-first interface with Framer Motion animations and dynamic product filtering.

SaaS Landing Page [Live Demo](#) | [Code](#)

Nov – Dec 2024

- Developed a multi-page marketing website using Next.js App Router.
- Created dynamic routes for blog posts and portfolio categories.
- Implemented dark and light theme switching using Context API.

Admin Dashboard Panel [Live Demo](#) | [Code](#)

Sep – Oct 2023

- Designed and developed an eight-page admin dashboard interface.
- Created pages for overview, projects, courses, file management, friends, pricing, profile, and settings.
- Built responsive layouts including statistics cards, tables, and navigation.

Roots Before React – Frontend Fundamentals [Live Demo](#) | [Code](#)

Jan – Jun 2023

- Developed five responsive landing pages using HTML5, CSS3, and JavaScript.
- Converted PSD designs into semantic and responsive websites.
- Applied BEM methodology, Flexbox, and CSS Grid.

ENGINEERING BACKGROUND

B.Sc. in Mechanical Power Engineering — Ain Shams University

2019 – 2024

- Graduation Project:** Solar Cell Cooling System — Designed an active thermal management system achieving a reported 15% improvement in photovoltaic panel efficiency.
- Academic Project:** Wind Turbine Pitch Control System — Designed and modeled dynamic pitch-control mechanisms and system logic.
- Engineering Internships:** National Authority for Tunnels — HVAC Systems; Arab Contractors — IC Engines.

EDUCATION & ADDITIONAL INFORMATION

Degree: B.Sc. Mechanical Power Engineering

University: Ain Shams University

Graduation: 2024

Military Service: Completed

Arabic: Native

English: Good